Is It Spacetime?

Discrete universes measured by counting: an invisible cosmological cutoff, a manifold detector, and quantum weights at $N = 200$

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Not for circulation. Claims are tagged established, literature or ours.

Abstract

We model spacetime as a discrete set of events with a causal order and nothing else, and ask two questions: how much geometry can be read off the order by counting alone, and which ways of producing such universes give something that looks like spacetime. First, Lorentz-invariant discreteness gives every event one neighbour per unit of rapidity, a known result. We take the finite size of the universe as the physical cutoff, which caps the rapidity at $\eta_{\max} \approx 142$ and singles out the cosmological frame. Applied to Sorkin's $1+1$ causal-set d'Alembertian, a finite past leaves no trace on plane waves of any wavelength. The averaged operator is a conditionally convergent integral, and an earlier "abrupt failure" we found came from cutting off rapidity instead of the past region. Second, we assemble a toolkit for $1+1$ dimensions that uses only counts. Proper time comes from the longest chain. Spatial distance comes from the causal overlaps of Boguñá and Krioukov, for which we find an exact $1{+}1$ identity, $d^2 = 2V_A V_B / V_C$, valid for any reference event in the common past; it recovers distances to 2.5–5%. In curved spacetime the same estimate drifts with the depth of the reference event as $1 - R\tau^2/16$, which we use to measure curvature. A Lorentzian multidimensional-scaling embedding then recovers flat positions to better than one event spacing and acts as a "manifold detector" through its eigenvalue spectrum, once a universe has at least 100–200 events. Third, we apply the detector. Universes grown event by event fail it. Universes weighted by $e^{-\beta S}$, with S the Benincasa–Dowker action, pass it in the continuum phase and fail it in the layered crystal phase. With quantum weights $e^{i\beta S}$, sampled from 10^6 universes at $N = 200$, universes with few layers are suppressed where classical weights favour them, as in exact sums at $N \leq 9$. In the reachable range ($\beta \lesssim 0.2$), however, the whole effect is explained by the spread of the action within each record ($P \propto n_r^2 e^{-\beta^2 \sigma_r^2}$), not by structured interference. Fourth, we ask whether the sum over histories can run as a film with a small state. Each birth needs only the past events with few events between it and them (a clock-free cut, error $\approx 0.06\,\sigma_S$ at every N), but these are spread over the whole age, and the set of events that later births still need is finite only in a closed universe without an event horizon. Gravity in $3+1$ dimensions and matter are still absent.

1 The question and the rules

In 1900 Planck assumed that energy comes in discrete quanta. We ask the analogous question for spacetime: can it be a discrete set of events that only *looks* continuous from far away, with general relativity (GR) as its large-scale description? The working constraints are:

C1 Discrete spacetime. Events and the causal order between them; the continuum is a large-scale limit.

C2 No background. No absolute space and no fixed aether; every statement is about relations between events.

C3 Non-locality regulated by the finite size of the universe (Sec. 2).

C4 Local Lorentz invariance for all boosts well below the cutoff.

C5 A cosmological clock computed from the universe's own state, e.g. the longest chain back to the beginning. It may be used as a bookkeeping clock for growth, but local laws must not depend on it, and predictions must not depend on which relational clock is used. Its frame should coincide with the matter (CMB) frame established; that is a test, not an input.

C6 Newton and Einstein from far away.

C7 Creation (and perhaps destruction) of events drives expansion, possibly through quantum creation operators.

C8 The links between events are quantum.

Missing so far is *matter*: nothing in the model yet plays the role of what gravitates.

A *fixed background* is structure the laws refer to but which nothing affects; C2 forbids it. A *dynamical preferred-frame field*, such as the Einstein-aether [1], breaks local Lorentz invariance. A *preferred frame of the state*, such as the CMB frame, comes from the contents of the universe while the laws stay invariant. C5 allows only the last kind.

The storyline. Section 2 explains why the finite size of the universe is our regulator. A discrete universe is only useful if it looks like spacetime, so Sec. 3 builds tools that answer "is it spacetime?" from the order alone, Sec. 4 applies them to three ways of producing universes: classical growth, $e^{-\beta S}$ weights and quantum weights, and Sec. 5 asks how much of the past a growing universe must remember.

2 Neighbours and the cosmological cutoff

One neighbour per unit of rapidity. literature Following causal set theory [2, 3], y is a *neighbour* (a *link*) of x if y is in the causal future of x and no other event lies causally between them. In $1+1$ dimensions, with density ρ, write a future event as $u = t - x = \tau e^{-\eta}$, $v = t + x = \tau e^{\eta}$, where τ is the proper time from the origin and η the rapidity. The area element is $dt\,dx = \tau\,d\tau\,d\eta$, and the interval up to the event has area $\tau^2/2$, so it is empty with probability $e^{-\rho\tau^2/2}$. Therefore

$$\langle N_{\text{links}} \rangle = \int d\eta \int_0^\infty \rho\,\tau\,e^{-\rho\tau^2/2}\,d\tau = \int d\eta \cdot 1. \tag{1}$$

There is one expected neighbour per unit of rapidity, so an infinite universe gives every event infinitely many neighbours. This is the elementary core of the Bombelli–Henson–Sorkin theorem [4]: Lorentz-invariant discreteness is non-local [5].

A finite universe caps the rapidity. ours A neighbour at proper time τ and rapidity η lies at distance $r \approx (\tau/2)e^{\eta}$. In a universe of size L this cannot exceed L, so

$$\eta_{\max} \simeq \ln \frac{2L}{\ell_{\text{P}}} \approx 142, \tag{2}$$

with $L \approx 4.4 \times 10^{26}\,\text{m}$ and $\ell_{\text{P}} = 1.6 \times 10^{-35}\,\text{m}$. Lorentz kinematics has been tested up to $\eta \approx 25$–27 (GZK protons and the highest-energy cosmic rays [6]), far below the cutoff. The only frame singled out is that of the universe as a whole, a frame of the state, as C5 requires. In $1+1$ dimensions an event has about $2\eta_{\max} \approx 280$ future neighbours. In $3+1$ dimensions the cutoff leaves $\sim (L/\ell_{\text{P}})^2 \approx 10^{124}$ (Appendix A lists the numerology we avoid).

Does the cutoff show up in wave propagation? literature Sorkin's 1+1 d'Alembertian [5] acts on a field by summing it over the past of x with layer weights. For density $\rho = \ell^{-2} = 1$ and averaged over sprinklings,

$$\bar{B}\phi(x) = 4\Big[-\tfrac{1}{2}\phi(x) + \int_{J^-(x)} d^2y\; e^{-V}\big(1 - 2V + \tfrac{1}{2}V^2\big)\phi(y)\Big], \tag{3}$$

where V is the volume of the interval between y and x. The three terms are links, next-to-links and next-to-next-to-links with weights $1, -2, 1$. On a plane wave $\phi = e^{-i(pu+qv)}$ it gives $4pq$, which is $\Box\phi/\phi$ in the convention used here (checked numerically).

ours The kernel integrates to *zero* per unit rapidity, $\int_0^\infty \tau\, d\tau\, e^{-\tau^2/2}(1 - \tau^2 + \tau^4/8) = 0$, but there are infinitely many units of rapidity. The integral in Eq. (3) is therefore only conditionally convergent: when ϕ does not oscillate along one light-cone direction, the answer depends on how the far past is cut off. We compared two cut-offs in the cosmological frame.

1. *A rapidity wedge*, $|\eta| \le \eta_{\max}$, with τ integrated to infinity. Write the past event as $(u, v) = -(\tau e^{-\eta}, \tau e^{\eta})$. Then $s = pe^{-\eta} + qe^{\eta}$ is the phase per unit τ, and the bracket becomes $-\frac{1}{2} + \int d\eta\, h(s)$, with h the τ-integral of the kernel times $e^{is\tau}$.
2. *A finite past region*, the diamond $0 \le a \le U$, $0 \le b \le U$ of displacements along u and v ($U \simeq L$). This is what a universe of finite age provides. The b-integral is elementary (incomplete gamma functions), and a is integrated numerically.

For a constant field, the diamond gives the bracket in closed form: $-\frac{1}{2} + \frac{1}{2}\big[1 - (1 - A)e^{-A}\big]$ with $A = U^2/2$, i.e. zero up to e^{-A}. The "$\frac{1}{2}$" is assembled at the two null edges of the region, $\frac{1}{4}$ each; the inner integral over the edge region reduces to digamma values, $-\frac{1}{2}\int_0^\infty e^{-x}(1 - 2x + \frac{1}{2}x^2)\ln x\, dx = \frac{1}{4}$. The results:

field	wedge	finite past	$\Box$
constant	-2	0	0
exact null wave ($p = 0$)	-1	0	0
massive, $pq = 10^{-9}$, a light-cone wavelength $> L$ ($L = e^{10}$)	-0.6 to -1.8	$4pq$	$4pq$

With an oscillating field the missing $\frac{1}{4}$ per null direction comes from the interior, at $u \sim 1/p$; with a non-oscillating one it comes from the edge of the past. The wedge has no edge, so it loses it. Aslanbeigi, Saravani and Sorkin [23] define the plane-wave eigenvalue only where the integral converges absolutely (complex momenta with a timelike, future-pointing imaginary part) and reach real momenta as a limit. Their $1+1$ result, $g(Z) = -Ze^{Z/2}E_2(Z/2)$ with $Z = p \cdot p/\rho$, goes smoothly to zero at null momentum, in agreement with the finite past.

Consequence. In $1+1$ the finite size of the universe leaves Sorkin's $\Box$ exactly continuum-like, up to $e^{-(\text{past volume})}$, for every wavelength. Through this operator the cutoff produces no dispersion correction and no change in the speed of light. That is good news for C3 and C4, which do not conflict here, and bad news for predictions. The $3+1$ case is not done.

3 Geometry by counting

Everything in this section uses only the causal order. To test the tools we generate universes by Poisson sprinkling at density ρ into a known spacetime, discard the coordinates, and compare what the tools find with the truth. We write $\ell = \rho^{-1/2}$ for the discreteness length in $1+1$ dimensions: a patch of volume ℓ^2 holds one event on average.

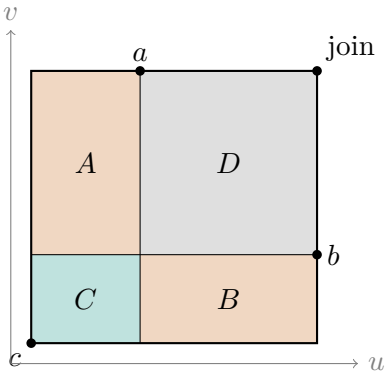


Figure 1: The rectangle identity in light-cone coordinates (rotate by 45° for the usual picture). The rectangle from c to the join of a and b splits into four tiles: the overlap C, the strips A and B seen by only one of a, b, and the diamond D with a and b at its corners. Opposite tiles have equal products, $C \cdot D = A \cdot B$, and D encodes the distance between a and b.

3.1 Time: the longest chain

literature The proper time between two events is estimated by the longest chain between them [7]. In a sprinkled 1+1 diamond this is the longest increasing subsequence of a random permutation, with mean $\simeq 2\sqrt{n}$ and Tracy–Widom fluctuations $\simeq 0.90\, n^{1/6}$ [8], so $\delta\tau/\tau \simeq 0.45\, n^{-1/3}$. Our simulations reproduce both (`sims/causal_diamond.py`). Equivalently $\tau^2 = 2V$, where the volume V of the interval is its number of events divided by ρ. ours (interpretation) Geometry measured by counting is intrinsically fuzzy. The fuzziness is a relation between events, not a smearing in a background space.

3.2 Space: causal overlaps and a rectangle identity

literature For two events a, b with neither before the other, Boguñá and Krioukov [9] pick an event c in their common past and compare the intervals $I(c, a)$ and $I(c, b)$:

$$A = I(c,a) \setminus I(c,b), \quad B = I(c,b) \setminus I(c,a), \quad C = I(c,a) \cap I(c,b), \qquad O = \frac{V_C}{\min(V_A, V_B) + V_C}. \tag{4}$$

With c chosen at equal proper time τ_c from a and b, in $1+1$ dimensions $d = \tau_c(1-O)/\sqrt{O}$. We re-derived this.

ours In $1+1$ dimensions the choice of c does not matter. Step by step (Fig. 1):

1. In light-cone coordinates $dt\, dx = du\, dv/2$, and for spacelike events $d^2 = \Delta x^2 - \Delta t^2 = -\Delta u\, \Delta v$.
2. With c at the origin, $I(c, a) = [0, u_a] \times [0, v_a]$ and $I(c, b) = [0, u_b] \times [0, v_b]$ are rectangles. Take $u_a < u_b$, $v_a > v_b$.
3. The rectangle from c to the join (u_b, v_a) splits into $C = [0, u_a] \times [0, v_b]$, $A = [0, u_a] \times [v_b, v_a]$, $B = [u_a, u_b] \times [0, v_b]$ and $D = [u_a, u_b] \times [v_b, v_a]$.
4. Opposite tiles have equal products: $C \cdot D = u_a v_b\, \Delta u\, \Delta v = A \cdot B$.
5. D has (u, v)-area $|\Delta u\, \Delta v| = d^2$, hence volume $d^2/2$.

Therefore, for *any* c in the common past,

$$d^2 = \frac{2\, V_A V_B}{V_C} \;\approx\; \frac{2\, N_A N_B}{\rho\,(N_C + 1)}. \tag{5}$$

N	$d < 2\ell$	2ℓ–10ℓ	$> 10\ell$
1000	23%	6.5%	5.2%
2000	15%	5.0%	3.4%
4000	19%	5.1%	2.5%

Table 1: Median relative error of Eq. (5) with the deepest c, flat $1+1$ sprinklings, 4 seeds (`sims/s4a_overlap_distance.py`). There is no significant bias above 2ℓ. The large-distance error falls roughly as $N^{-1/2}$. Below 2ℓ the error stays at $\sim 20\%$: this is the discreteness scale.

The strips A and B are long and thin, so they hold many events even when $d \sim \ell$; this is also why the Boguñá–Krioukov estimator works at short distances. Their equal-time choice is the special case $V_A = V_B$. The identity holds only in $1+1$ dimensions, because in more dimensions the overlap of two diamonds is not a diamond. The counts N_A, N_B, N_C come from disjoint regions and are therefore independent Poisson variables. $N_A N_B$ is then unbiased, but $1/N_C$ is not; $1/(N_C+1)$ has mean $(1-e^{-\lambda})/\lambda$ and removes the bias. Before this fix, a flat control read $+11\%$.

3.3 Curvature: the estimate drifts with depth

literature Curvature can be read from counts through the Benincasa–Dowker operator [10, 11]. In our runs (session 2) it gave the right sign and order of magnitude for de Sitter, but was noisy at 10^4 events. Related continuum results give the curvature corrections to the volume of a small causal diamond [12] and to chain counts inside it [13].

ours The rectangle identity provides a second route. Two-dimensional de Sitter space, $ds^2 = (H\eta)^{-2}(-d\eta^2 + dx^2)$, has the same light cones as flat space, so A, B, C, D are the same rectangles, but volumes are weighted by $dV = 2\,du\,dv/(H^2(u+v)^2)$. Integrating over a rectangle gives a logarithm of the conformal times at its corners,

$$V_\square = \frac{2}{H^2} \ln \frac{\eta_L\, \eta_R}{\eta_{\text{bottom}}\, \eta_{\text{top}}}, \tag{6}$$

and $C \cdot D = A \cdot B$ no longer holds exactly. For a, b on one slice and c a proper time τ_c below them, expanding for small d and then small τ_c gives

$$\frac{d^2_{\text{est}}}{d^2} = 1 - \frac{(H\tau_c)^2}{8} + \cdots = 1 - \frac{R\,\tau_c^2}{16} + \ldots, \qquad R = 2H^2. \tag{7}$$

In flat space every c gives the same answer. In curved space the answer depends on how deep c lies, and the size of that drift is the curvature. We checked the coefficient $1/8$ numerically from the exact volumes (0.1252 at small τ). In sprinklings ($\rho = 10^4$, test events inserted by hand 14ℓ apart), the measured ratio agrees with the continuum in every depth bin for 200 fresh seeds, and a flat control shows no drift. Fitting $d^2_{\text{est}} = \alpha + \beta\tau^2$ with τ also from counts gives $R = -16\beta/\alpha$:

ρ	flat (true $R = 0$)	de Sitter (true $R = 2$; continuum fit in this window 2.13)
4000	-0.7 ± 0.6	1.3 ± 0.7
10^4	0.3 ± 0.4	1.6 ± 0.4

It is right on average, but a single pair is useless: the run-to-run spread is about 4, so about 100 pairs are needed for ± 0.4. Three web searches found no curvature version of the overlap estimator, so it is ours in the sense "not found in a brief search".

3.4 Shape: an embedding that doubles as a manifold detector

literature Given squared intervals between M events, classical multidimensional scaling (MDS) with a Lorentzian Gram matrix reconstructs coordinates (cf. [14]). With $s_{ij} = (x_i - x_j)^T \eta (x_i - x_j)$, double centring $G = -\frac{1}{2} JSJ$, $J = I - \mathbf{1}\mathbf{1}^T/M$, gives $G = X\eta X^T$. A $1+1$ Lorentzian set shows one large negative eigenvalue (time) and one large positive one (space), with the rest near zero, and the eigenvectors give coordinates up to a Lorentz transformation.

ours With $s = -2N(I)/\rho$ for timelike pairs and Eq. (5) for spacelike pairs (all from counts), on $M = 150$ events of a flat sprinkling ($N = 2000$):

- positions are recovered to $0.43 \pm 0.07\,\ell$ across a region about 38ℓ wide, better than one event spacing;
- the map to the true coordinates is a Lorentz transformation to within 3–10%;
- the spectrum is $\lambda_- \approx -2.2$, $\lambda_+ \approx 2.2$, with all others below 0.1;
- a $2+1$ sprinkling fed through the same $1+1$ pipeline shows *two* large positive eigenvalues: the spectrum counts spatial dimensions even with the wrong distance formula.

We turn this into a detector with four intrinsic numbers: w_2, the share of the spectrum in one time and one space direction; $r_\pm$, the ratio of the second to the first eigenvalue of each sign; and the *stress*, the rms misfit of the best $1+1$ Minkowski embedding. Two calibrations matter. (i) Size: on flat universes, stress and r_+ are 0.22/0.42 at $N = 20$, 0.33/0.41 at 50, 0.20/0.16 at 100, 0.11/0.08 at 200 and 0.08/0.05 at 400. The detector is reliable only from $N \approx 100$–200; below 50 a flat universe looks like a crystal. (ii) Curvature: on de Sitter diamonds of fixed $N = 400$, the stress stays at the flat value up to $H\tau \approx 1$ and rises to 0.21 at $H\tau = 4$. So the embedding answers "is it a sheet?", while Eq. (7) measures curvature.

4 Which universes are spacetime?

4.1 Classical growth fails

literature In sequential growth [15] events are born one at a time, and each newborn picks causes among existing events. ours We ran two rules: fixed link probability p (a steady state of fixed width) and $p_n = 3/n$ (an expanding universe from a single first event; layers of $1, 2, 3, 3, 5, 7, \ldots, \sim 280$ events). The second rule uses the birth order as a hidden clock, which C2 forbids; the covariant route is through cosmological renormalisation of p [16, 17]. Neither rule gives a $1+1$ spacetime. Their Myrheim–Meyer dimension drifts with scale, falling to ~ 1.2 for fixed p (a thin tube) and rising from 1.6 to 2.8 for $p = 3/n$. On the detector at $N = 100$ they fail every number: $w_2 \approx 0.33$ against 0.65 for the calibration, and stress ≈ 0.65 against 0.19.

4.2 Weights $e^{-\beta S}$: a continuum phase and a crystal

literature Weighting $1+1$ orders by $e^{-\beta S}$, with S the Benincasa–Dowker action, gives a phase transition between a continuum-like phase and a layered "crystal" [18, 19, 20]. ours We reproduce it ($\beta_c \approx 1.2, 0.7, 0.5$ for $N = 30, 50, 70$, at non-locality parameter $\epsilon = 0.21$). An $O(N^2)$ Metropolis update (a swap changes only two rows and columns of the causal matrix, so $K' = CC + C'E + EC$ with thin E) brings $N = 200$ to 0.03 s per sweep. Two clocks, height and past volume, give the same cosmic ordering in both phases, except for pairs within one or two layers (C5).

$N = 200$	ordering fraction	layers	w_2	r_+	stress
flat calibration	0.50	24	0.75	0.07	0.10
$\beta = 0.05$ (continuum)	0.51	24	0.74	0.10	0.11
$\beta = 0.6$ (crystal)	0.59	9	0.82	**0.39**	**0.28**

The continuum phase is indistinguishable from flat spacetime. The crystal asks for an extra spatial direction and fits badly.

$N = 400$: hysteresis. ours (preliminary) At $N = 400$, $\beta = 0.3$ did not crystallise from a random start. To tell slow equilibration from a shifted transition, we start each chain either hot (random) or cold (a two-layer order, the lowest-action layered state we tried). On the laptop (100 sweeps) the two-layer start keeps $S/N \approx -34$ for $\beta \gtrsim 0.04$, while hot chains reach at most $S/N \approx -6$ (an S3a-type crystal, $r \approx 0.6$) up to $\beta = 1$. Hot and cold chains sit in different states over a wide range of β, so the S3a "crystal" may be a metastable intermediate. A run of 80 chains with 1.5×10^4 sweeps each is in progress on the cluster.

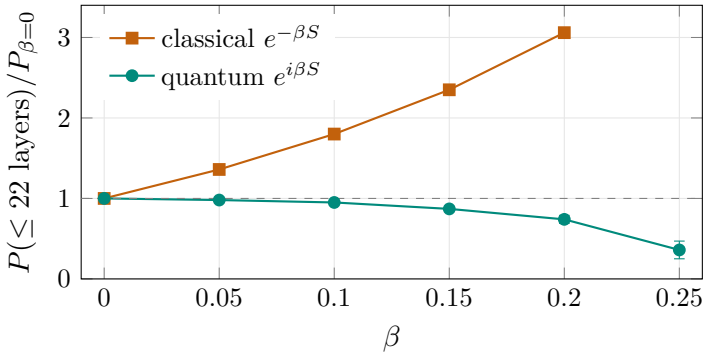
4.3 Quantum weights $e^{i\beta S}$

Records. ours (lesson from session 2) Phases only matter if different histories end in the same outcome. If every universe is its own outcome, $|e^{i\beta S}|^2 = 1$ and nothing interferes. We therefore choose a *record*, a relational property shared by many universes, and let universes with the same record interfere: $P_q(r) = |A_r|^2 / \sum_{r'} |A_{r'}|^2$, $A_r = \sum_{\text{record } r} e^{i\beta S}$.

A bound. ours (likely known) Since $|e^{i\beta S}| = 1$, $|A_r|$ is at most the number of universes with record r. In the quantum sum, rare universes such as crystals can only win if the common, manifold-like ones cancel each other out. With $e^{-\beta S}$, rare universes with low action win by sheer weight.

Exact sums, $N \leq 9$. ours Summing over all $N!$ orders with record = number of layers, $e^{-\beta S}$ collapses onto flat layered universes ($N = 9$, $\beta = 8$: 2 layers with 76%), while $e^{i\beta S}$ suppresses them by destructive interference (2 layers: 0%; 5 layers: 62%). Such universes are too small to say whether they are spacetime.

Sampling, $N = 200$. ours The amplitude of a record is proportional to the average of $e^{i\beta S}\mathbf{1}_r$ over uniformly random orders, which can be sampled. This works while the average phase, $\approx e^{-\beta^2\sigma_S^2/2}$, is not tiny. At $N = 200$, $\sigma_S = 12.0$, so the average phase is 0.49 at $\beta = 0.1$ and 0.06 at $\beta = 0.2$. We drew 10^6 orders on the cluster (50 independent seeds; errors are jackknife over seeds). The record is the layer count (the height clock of C5), grouped into bins. The probability of the few-layer bin (≤ 22 layers; the mean is 24.6), relative to its value at $\beta = 0$:



Classically, raising β drives the universe toward few layers, the approach to the crystal. Quantum weights do the opposite and suppress few-layer universes. This agrees with the exact sums and with Loomis and Carlip [24], who show analytically that e^{iS} suppresses two-layer orders. (The classical reweighting is reliable up to $\beta = 0.2$, with an effective sample size of 1,022.)

But the mechanism is simple. Within each record the action is close to Gaussian, with spread σ_r. Then $|A_r| \approx n_r e^{-\beta^2\sigma_r^2/2}$, so

$$P_q(r) \propto n_r^2\, e^{-\beta^2\sigma_r^2}, \tag{8}$$

and this reproduces the sampled numbers to 0.001 for $\beta \leq 0.15$. Few-layer universes have a slightly larger spread ($\sigma = 11.73$ against 11.48), so their phases cancel more. In the range that sampling can reach, "quantum suppression" is a variance effect, not a structured interference pattern. With record = stress quartile, quantum and classical weights drift by the same small amount (Q1: 0.261 ± 0.003 against 0.258 at $\beta = 0.15$). A 1,600-sample pilot that suggested a large shift toward manifold-like universes was noise. Uniform sampling is exhausted at $\beta \approx 0.2$ for $N = 200$; real interference structure needs exact or layer-by-layer methods.

5 Growing it as a film: how much past does the present need?

A cosmological clock (C5) suggests rewriting the sum over histories as a film: a state for "now" and an operator taking it to the next moment (C7). That needs the action to split into steps that each see only a small piece of the past. We measured how far that goes.

Thick height layers fail. ours Truncate S to pairs at most m height layers apart. Links skip layers: in continuum $1+1$ orders, 61% of links skip at least one layer at $N = 200$, and the longest spans 23 of about 24 layers. To keep the error below $0.25\,\sigma_S$ the memory must be $m \approx 9, 10, \gtrsim 14$ layers at $N = 100, 200, 400$, about 40% of the age. At $N \leq 9$ the truncated quantum sums deviate from the exact ones by up to 0.47 (total variation) when only 4.5% of the pairs are dropped. A volume clock gives different answers from the height clock, so a height truncation also breaks C5. (The crystal needs only $m \approx 4$.)

Growth and causal closeness. literature We let events be born one at a time, each with any set of existing events as its past but never in the past of an existing event: Rideout–Sorkin's internal temporality [15]. Every causal set can be grown this way, the dates (heights) of events never change, and $e^{i\beta S}$ does not depend on the birth order. ours A birth f changes the action by

$$\Delta S_f = 4\epsilon - 8\epsilon^2 \sum_{x<f} f(n_{xf}, \epsilon), \tag{9}$$

with n_{xf} the number of events between x and f (summing over births reproduces S to 10^{-10}). Keeping only x with $n_{xf} \leq 10$ leaves an error of 0.06–0.07 σ_S at $N = 100, 200, 400$, flat in N, and 0.07 in the crystal. The cut is a count, so it uses no clock. About 16, 22, 29 past events are kept per birth, growing like $\ln N$, but they span the whole age: the influence of the past is sparse and long-range in time, and local in "events in between".

The thick present. ours If f needs x, it also needs everything between them, so a state holding all events that some later birth still needs is self-contained. We measured its size in hindsight, which gives a lower bound:

universe $(1+1,\ n_{\text{cut}}=10)$	state
diamond (open space)	0.6–$0.8\,N$
static circle of length W	$\approx 0.5\,W^2$, independent of the age
decelerating closed FRW ($a \propto t^{1/2}$)	oldest needed event $\approx 0.5\,W$ of conformal time back
accelerating closed FRW (e^{Ht}, t^2)	0.9–1.0 of the past, never retired

An event stays needed until its light cone wraps around space, about $W/2$ of conformal time. The film therefore has a finite memory if and only if the universe has no event horizon. Our universe accelerates, so by this measure its present needs the whole past. The per-birth step is cheap; the storage is not.

6 Discussion

Where this sits. The kinematics is causal set theory: links, infinite valency, Number = Volume, the longest chain as proper time, growth, the Benincasa–Dowker action, the phase transition and the overlap distances are theirs. Our additions are (i) the finite size of the universe as the regulator of the non-locality, with the cosmological frame accepted as a frame of the state; (ii) the rectangle identity (5); (iii) curvature from the depth drift (7); (iv) a calibrated, intrinsic manifold detector; (v) quantum record probabilities by sampling at $N = 200$, with the $|A_r|$ bound and the variance formula $P_q \propto n_r^2 e^{-\beta^2\sigma_r^2}$; and (vi) the check that a finite past keeps Sorkin's $\Box$ exact in $1+1$. Items (ii)–(iv) were not found in brief searches. (v) confirms a known trend at larger N and explains it. (vi) agrees with the analytic treatment of [23]; what we add is the finite universe as the physical regulator. (vii) The causal-closeness cut of the growth step and the thick-present rule (memory $\approx W/2$ of conformal time; finite iff no event horizon) were not looked for in the literature beyond Rideout–Sorkin's growth rule, so they are ours in the weakest sense: not searched. On the dynamics side, a cosmological clock (C5) would license writing the sum over histories as a transfer matrix between cosmic-time layers, with operators that create events (C7). That puts us next to causal dynamical triangulations and Hořava–Lifshitz gravity, whose preferred foliation is fundamental rather than emergent; we must show that the difference is real.

Limitations.

1. *Everything is* $1+1$. The rectangle identity is $1+1$ only, and $1+1$ GR has no Newtonian limit, so C6 needs $3+1$.
2. $3+1$ *is not tamed*: 10^{124} neighbours per event.
3. *No matter.* Nothing gravitates.
4. *Small universes.* The detector needs $N \gtrsim 100$–200. Exact quantum sums stop at $N \approx 12$, and sampled ones are limited by the average phase.
5. *The quantum result is a variance effect.* Sampling reaches only $\beta \lesssim 0.2$ at $N = 200$. The record and ϵ are chosen, not derived.
6. *No prediction.* In $1+1$ the cutoff is invisible to wave propagation, not merely far beyond tests.
7. *The film needs the whole past* in any accelerating universe. If the universe is to be an efficient process, something must manage that memory [26]; we have not found it.

Next steps.

- Quantum growth with the whole past as state and the clock-free per-birth step: amplitudes on the choice of each newborn's past (C8), and the counting measure, not only the amplitude.

- How much information does the thick present really carry? Compare with area (holographic) bounds rather than counting events.
- Finish the $N = 400$ hot/cold runs: the location of the transition and whether the two-layer state is the true crystal.
- The cutoff check in $3 + 1$.
- The general Boguñá–Krioukov estimator and the detector in $2 + 1$ and $3 + 1$.

A Paths explored and dropped

- **A regular crystal of events.** Polygonal light cones and a lattice rest frame [21, 22].
- **Quantum blur of positions.** Blur in *what*? It presupposes a background, and an equal superposition over rapidity cannot be normalised.
- **A quantum chain as the source of blur.** Background-dependent. The atomic-linewidth analogy was overstated: coupling *reduces* the blur.
- **Expansion as a source and gravity as a sink of points** (river model [27]). It needs an aether.
- **A link to Bell non-locality.** Links are timelike; Bell correlations are spacelike.
- $(L/\ell_{\mathrm{P}})^2 \sim 10^{122}$ **and the cosmological constant.** Numerology: every combination of L and ℓ_{P} is a power of 10^{61}.
- **The manifold detector on exact quantum sums.** At $N \leq 12$ the detector cannot tell a flat universe from a crystal; replaced by sampling.
- $\eta_{\max}$ **as a dispersion correction.** Killed in $1+1$ (Sec. 2). An intermediate version claimed a jump to $-1/\ell^2$ once a wavelength exceeds L; that was an artefact of a rapidity-wedge regulator.
- $\eta_{\max}$ **as the long-link suppression of Caravelli and Markopoulou [25].** A different object: a spatial lattice with external time, giving an energy-dependent mass.
- **A transfer matrix on thick height layers.** The memory must grow to $\sim 40\%$ of the age, and the truncation breaks clock independence (Sec. 5).
- **A finite-state film in an accelerating universe.** The thick present never retires its events (Sec. 5).
- **Quantum weights at** $N = 100$ **on stress quartiles.** Superseded by $N = 200$; the small-sample hint was noise.

References

[1] T. Jacobson, D. Mattingly, "Gravity with a dynamical preferred frame", Phys. Rev. D 64, 024028 (2001), gr-qc/0007031.

[2] L. Bombelli, J. Lee, D. Meyer, R. D. Sorkin, "Space-time as a causal set", Phys. Rev. Lett. 59, 521 (1987).

[3] S. Surya, "The causal set approach to quantum gravity", Living Rev. Rel. (2019), arXiv:1903.11544.

[4] L. Bombelli, J. Henson, R. D. Sorkin, "Discreteness without symmetry breaking: a theorem", gr-qc/0605006.

[5] R. D. Sorkin, "Does locality fail at intermediate length-scales?", gr-qc/0703099.

[6] S. Coleman, S. L. Glashow, "High-energy tests of Lorentz invariance", Phys. Rev. D 59, 116008 (1999), hep-ph/9812418.

[7] G. Brightwell, R. Gregory, "Structure of random discrete spacetime", Phys. Rev. Lett. 66, 260 (1991).

[8] J. Baik, P. Deift, K. Johansson, "On the distribution of the length of the longest increasing subsequence of random permutations", J. Amer. Math. Soc. 12, 1119 (1999).

[9] M. Boguñá, D. Krioukov, "Measuring spatial distances in causal sets via causal overlaps", Phys. Rev. D 110, 024008 (2024), arXiv:2401.17376.

[10] D. M. T. Benincasa, F. Dowker, "The scalar curvature of a causal set", Phys. Rev. Lett. 104, 181301 (2010).

[11] F. Dowker, L. Glaser, "Causal set d'Alembertians for various dimensions", arXiv:1305.2588.

[12] G. W. Gibbons, S. N. Solodukhin, "The geometry of small causal diamonds", Phys. Lett. B 649, 317 (2007), hep-th/0703098.

[13] M. Roy, D. Sinha, S. Surya, "The discrete geometry of a small causal diamond", Phys. Rev. D 87, 044046 (2013), arXiv:1212.0631.

[14] S. Johnston, "Simpler embeddings of causal sets into Minkowski spacetime", arXiv:2502.09701.

[15] D. P. Rideout, R. D. Sorkin, "A classical sequential growth dynamics for causal sets", Phys. Rev. D 61, 024002 (2000).

[16] X. Martín, D. O'Connor, D. P. Rideout, R. D. Sorkin, "On the renormalization transformations induced by cycles of expansion and contraction in causal set cosmology", Phys. Rev. D 63, 084026 (2001).

[17] M. Ahmed, D. Rideout, "Indications of de Sitter spacetime from classical sequential growth dynamics of causal sets", Phys. Rev. D 81, 083528 (2010).

[18] S. Surya, "Evidence for a phase transition in 2D causal set quantum gravity", arXiv:1110.6244.

[19] G. Brightwell, J. Henson, S. Surya, "A 2D model of causal set quantum gravity: the emergence of the continuum", arXiv:0706.0375.

[20] L. Glaser, D. O'Connor, S. Surya, "Finite size scaling in 2d causal set quantum gravity", Class. Quant. Grav. (2018), arXiv:1706.06432.

[21] D. Burago, "Periodic metrics"; statement in arXiv:1104.4460.

[22] H. Weyl, *Philosophy of Mathematics and Natural Science* (1949).

[23] S. Aslanbeigi, M. Saravani, R. D. Sorkin, "Generalized causal set d'Alembertians", JHEP 06 (2014) 024, arXiv:1403.1622.

[24] S. P. Loomis, S. Carlip, "Suppression of non-manifold-like sets in the causal set path integral", arXiv:1709.00064.

[25] F. Caravelli, F. Markopoulou, "Disordered locality and Lorentz dispersion relations", Phys. Rev. D 86, 024019 (2012), arXiv:1201.3206.

[26] S. Lloyd, "Computational capacity of the universe", Phys. Rev. Lett. 88, 237901 (2002), quant-ph/0110141.

[27] A. J. S. Hamilton, J. P. Lisle, "The river model of black holes", Am. J. Phys. 76, 519 (2008).